

数据重构

2026 年 5 月 21 日

1 数据重构

```
[1]: # 加载所需的库
# 如果出现 ModuleNotFoundError: No module named 'xxxx'
# 你只需要在终端/cmd 下 pip install xxxx 即可
%matplotlib inline
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
[2]: # 载入 data 文件中的:train-left-up.csv
text = pd.read_csv('/home/jovyan/work/datasets/6645a647c4db0a94029f0bf5-momodel/
↳dataset/train-left-up.csv')
text.head()
```

```
[2]:
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name
0	Braund, Mr. Owen Harris
1	Cumings, Mrs. John Bradley (Florence Briggs Th...
2	Heikkinen, Miss. Laina
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)
4	Allen, Mr. William Henry

1.1 1. 数据的合并

任务一：将 data 文件夹里面的所有数据都载入，与之前的原始数据相比，观察他们的之间的关系

```
[3]: text_left_up = pd.read_csv("/home/jovyan/work/datasets/
↳6645a647c4db0a94029f0bf5-momodel/dataset/train-left-up.csv")
text_left_down = pd.read_csv("/home/jovyan/work/datasets/
↳6645a647c4db0a94029f0bf5-momodel/dataset/train-left-down.csv")
text_right_up = pd.read_csv("/home/jovyan/work/datasets/
↳6645a647c4db0a94029f0bf5-momodel/dataset/train-right-up.csv")
text_right_down = pd.read_csv("/home/jovyan/work/datasets/
↳6645a647c4db0a94029f0bf5-momodel/dataset/train-right-down.csv")
```

```
[5]: text_left_up.head()
```

```
[5]:   PassengerId  Survived  Pclass  \
0             1         0        3
1             2         1        1
2             3         1        3
3             4         1        1
4             5         0        3
```

```
                                Name
0                        Braund, Mr. Owen Harris
1  Cumings, Mrs. John Bradley (Florence Briggs Th...
2                        Heikkinen, Miss. Laina
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)
4                        Allen, Mr. William Henry
```

```
[6]: text_left_down.head()
```

```
[6]:   PassengerId  Survived  Pclass                                Name
0          440         0        2      Kvillner, Mr. Johan Henrik Johannesson
1          441         1        2  Hart, Mrs. Benjamin (Esther Ada Bloomfield)
2          442         0        3                        Hampe, Mr. Leon
3          443         0        3      Petterson, Mr. Johan Emil
4          444         1        2      Reynaldo, Ms. Encarnacion
```

```
[7]: text_right_down.head()
```

```
[7]:      Sex  Age  SibSp  Parch      Ticket    Fare Cabin Embarked
0   male  31.0    0     0   C.A. 18723   10.500   NaN      S
1  female  45.0    1     1  F.C.C. 13529   26.250   NaN      S
2   male  20.0    0     0    345769    9.500   NaN      S
3   male  25.0    1     0    347076    7.775   NaN      S
4  female  28.0    0     0    230434   13.000   NaN      S
```

```
[8]: text_right_up.head()
```

```
[8]:      Sex  Age  SibSp  Parch      Ticket    Fare Cabin Embarked
0   male  22.0    1     0    A/5 21171    7.2500   NaN      S
1  female  38.0    1     0    PC 17599   71.2833   C85      C
2  female  26.0    0     0  STON/O2. 3101282    7.9250   NaN      S
3  female  35.0    1     0    113803   53.1000  C123      S
4   male  35.0    0     0    373450    8.0500   NaN      S
```

【提示】结合之前我们加载的 train.csv 数据，大致预测一下上面的数据是什么

任务二：使用 concat 方法：将数据 train-left-up.csv 和 train-right-up.csv 横向合并为一张表，并保存这张表为 result_up

```
[9]: list_up = [text_left_up, text_right_up]
result_up = pd.concat(list_up, axis=1)
result_up.head()
```

```
[9]:      PassengerId  Survived  Pclass  \
0                1         0        3
1                2         1        1
2                3         1        3
3                4         1        1
4                5         0        3

                                     Name      Sex  Age  SibSp  \
0                Braund, Mr. Owen Harris   male  22.0    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0    1
2                Heikkinen, Miss. Laina   female  26.0    0
```

```

3      Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                                Allen, Mr. William Henry    male  35.0      0

```

```

      Parch      Ticket    Fare Cabin Embarked
0         0   A/5 21171    7.2500   NaN        S
1         0     PC 17599   71.2833   C85        C
2         0 STON/O2. 3101282    7.9250   NaN        S
3         0      113803   53.1000  C123        S
4         0      373450    8.0500   NaN        S

```

任务三：使用 `concat` 方法：将 `train-left-down` 和 `train-right-down` 横向合并为一张表，并保存这张表为 `result_down`。然后将上边的 `result_up` 和 `result_down` 纵向合并为 `result`。

```

[10]: list_down=[text_left_down,text_right_down]
      result_down = pd.concat(list_down,axis=1)
      result = pd.concat([result_up,result_down])
      result.head()

```

```

[10]:   PassengerId  Survived  Pclass  \
0             1         0        3
1             2         1        1
2             3         1        3
3             4         1        1
4             5         0        3

```

```

                                Name      Sex  Age  SibSp  \
0                Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                Heikkinen, Miss. Laina    female  26.0      0
3      Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                Allen, Mr. William Henry    male  35.0      0

```

```

      Parch      Ticket    Fare Cabin Embarked
0         0   A/5 21171    7.2500   NaN        S
1         0     PC 17599   71.2833   C85        C
2         0 STON/O2. 3101282    7.9250   NaN        S
3         0      113803   53.1000  C123        S

```

```
4      0      373450    8.0500    NaN      S
```

任务四：使用 DataFrame 自带的方法 `join` 和 `append`：完成任务二和任务三的任务

```
[11]: resul_up = text_left_up.join(text_right_up)
      result_down = text_left_down.join(text_right_down)
      result = result_up.append(result_down)
      result.head()
```

```
[11]: PassengerId  Survived  Pclass  \
0             1           0         3
1             2           1         1
2             3           1         3
3             4           1         1
4             5           0         3

                                     Name    Sex  Age  SibSp  \
0                               Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                               Heikkinen, Miss. Laina  female  26.0      0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0      1
4                               Allen, Mr. William Henry    male  35.0      0

      Parch      Ticket    Fare Cabin Embarked
0         0    A/5 21171    7.2500   NaN        S
1         0      PC 17599   71.2833   C85        C
2         0  STON/O2. 3101282    7.9250   NaN        S
3         0      113803   53.1000  C123        S
4         0      373450    8.0500   NaN        S
```

任务五：使用 Pandas 的 `merge` 方法和 DataFrame 的 `append` 方法：完成任务二和任务三的任务

```
[12]: result_up = pd.
      ↪merge(text_left_up,text_right_up,left_index=True,right_index=True)
      result_down = pd.
      ↪merge(text_left_down,text_right_down,left_index=True,right_index=True)
      result = resul_up.append(result_down)
```

```
result.head()
```

```
[12]:
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

【思考】对比 `merge`、`join` 以及 `concat` 的方法的不同以及相同。思考一下在任务四和任务五的情况下，为什么都要求使用 `DataFrame` 的 `append` 方法，如何只要求使用 `merge` 或者 `join` 可不可以完成任务四和任务五呢？

完成的数据保存为 `result.csv`

```
[14]: result.to_csv('/home/jovyan/work/results/result.csv')
```

换一种角度看数据：

任务六：将我们的数据变为 `Series` 类型的数据

【思考】`stack` 函数是干什么的？

```
[15]: # 将完整的数据加载出来
text = pd.read_csv('/home/jovyan/work/results/result.csv')
```

```
text.head()
# 代码写在这里
unit_result=text.stack().head(20)
unit_result.head()
```

```
[15]: 0  Unnamed: 0          0
      PassengerId      1
      Survived        0
      Pclass          3
      Name      Braund, Mr. Owen Harris
dtype: object
```

```
[17]: # 将代码保存为 unit_result.csv
unit_result.to_csv('/home/jovyan/work/results/unit_result.csv')
```

```
[18]: test = pd.read_csv('/home/jovyan/work/results/unit_result.csv')
```

```
[19]: test.head()
```

```
[19]:  Unnamed: 0  Unnamed: 1          0
0          0  Unnamed: 0          0
1          0  PassengerId      1
2          0    Survived        0
3          0    Pclass          3
4          0      Name  Braund, Mr. Owen Harris
```

1.2 2. 数据聚合与运算

任务一：通过《Python for Data Analysis》P303、Google or Baidu 来学习了解 GroupBy 机制
在了解 GroupBy 机制之后，运用这个机制完成一系列的操作，来达到我们的目的。

下面通过几个任务来熟悉 GroupBy 机制。

任务二：计算泰坦尼克号男性与女性的平均票价

```
[20]: df = text['Fare'].groupby(text['Sex'])
      means = df.mean()
      means
```

```
[20]: Sex
      female    44.479818
      male      25.523893
      Name: Fare, dtype: float64
```

任务三：统计泰坦尼克号中男女的存活人数

```
[21]: survived_sex = text['Survived'].groupby(text['Sex']).sum()
      survived_sex.head()
```

```
[21]: Sex
      female    233
      male      109
      Name: Survived, dtype: int64
```

任务四：计算客舱不同等级的存活人数

```
[22]: survived_pclass = text['Survived'].groupby(text['Pclass'])
      survived_pclass.sum()
```

```
[22]: Pclass
      1    136
      2     87
      3    119
      Name: Survived, dtype: int64
```

任务五：统计在不同等级的票中的不同年龄的船票花费的平均值

```
[23]: text.groupby(['Pclass', 'Age'])['Fare'].mean().head()
```

```
[23]: Pclass  Age
      1      0.92    151.5500
           2.00    151.5500
           4.00     81.8583
          11.00    120.0000
          14.00    120.0000
      Name: Fare, dtype: float64
```

任务六：将任务二和任务三的数据合并，并保存到 sex_fare_survived.csv

```
[26]: result = pd.merge(means,survived_sex,on='Sex')
result.to_csv('/home/jovyan/work/results/sex_fare_survived.csv')
result
```

```
[26]:           Fare  Survived
Sex
female  44.479818      233
male    25.523893      109
```

任务七：得出不同年龄的总的存活人数，然后找出存活人数最多的年龄段，最后计算存活人数最高的存活率（存活人数/总人数）

```
[27]: # 不同年龄的存活人数
survived_age = text['Survived'].groupby(text['Age']).sum()
survived_age.head()
```

```
[27]: Age
0.42    1
0.67    1
0.75    2
0.83    2
0.92    1
Name: Survived, dtype: int64
```

```
[28]: # 找出最大值的年龄段
survived_age[survived_age.values==survived_age.max()]
```

```
[28]: Age
24.0    15
Name: Survived, dtype: int64
```

```
[29]: _sum = text['Survived'].sum()
print(_sum)
```

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```
[30]: # 首先计算总人数
_sum = text['Survived'].sum()
```

```
print("sum of person:"+str(_sum))

precetn =survived_age.max()/_sum

print("最大存活率: "+str(precetn))
```

sum of person:342

最大存活率: 0.043859649122807015